

Week 7: Dump-Bed Floor Hand Structural Analysis

Lunar Regolith Transport Rover

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Summer 2026 Conceptual Design

Primary component	6061-T6 aluminum dump-bed floor
Analysis status	Preliminary conceptual hand calculation



Figure 1: Week 7 dump-bed floor analysis region.

1 Week 7 Objective

The Week 7 objective is to perform a transparent hand structural analysis of one rover component before using finite-element analysis (FEA). The analysis checks load, pressure, shear, bending moment, bending stress, elastic deflection, and factor of safety. It also identifies a practical floor thickness and cross-member spacing for the next CAD revision.

The rover is a semi-autonomous six-wheel vehicle that transports loose lunar regolith from a collection area to a habitat-shielding area. It does not excavate the regolith. The rear-hinged dump bed is loaded by astronauts or a separate excavation rover and is raised by a linear actuator for unloading, consistent with the Week 3 mission definition [1].

2 Component Selected for Analysis

The dump-bed floor is selected as the primary Week 7 component. The floor directly supports the known 340 kg payload and has known plan dimensions established in Week 4 [2]. It can therefore be reduced to a controlled beam-strip model with assumptions that can be stated and checked.

3 Why This Component Was Selected

The dump bed is the stronger and more manageable choice for the Week 7 hand analysis. Its load magnitude and loaded area are known, and its supports can be represented by cross-members. The chassis would eventually provide a more system-level result, but its final rail section, battery position, actuator mount, bed hinge reactions, suspension reaction locations, and concentrated equipment loads are not locked. A chassis calculation made before those inputs are fixed could look precise while using the wrong load path.

The chassis is still meaningful because it carries the complete rover and connects the suspension reactions. A first chassis model can use two longitudinal rails as simply supported beams over an approximate 2.2 m wheelbase, with half the rover weight applied to each rail. A brief chassis sanity check is included in Section 16, but it is not treated as a finalized chassis design.

This component choice also follows the Week 3 load path: regolith payload to dump bed, dump bed to main chassis, chassis to rocker-pivot brackets, and then into the rocker-bogie arms and wheels [1]. The dump bed is therefore mounted to the main chassis rather than directly to moving suspension members. The Week 7 floor calculation addresses the first structural element in that load path while leaving the hinge, chassis, pivot brackets, and suspension reactions outside the present floor-sheet model.

4 Geometry and Material Assumptions

4.1 Geometry and payload

The payload, bed dimensions, and rover masses in Table 1 are carried forward from Weeks 4 and 5 [2, 3]. The lunar gravitational acceleration is the NASA average surface value [4].

Table 1: Locked geometry, payload, environment, and design inputs.

Quantity	Symbol	Value
Payload mass	m_p	340 kg
Bed internal length	L_{bed}	1.20 m
Bed internal width	B_{bed}	0.75 m
Bed internal height	H_{bed}	0.35 m
Floor area	A	0.90 m^2
Internal volume	V_{bed}	0.315 m^3
Implied loose-regolith density	ρ_{reg}	$1079 \text{ kg/m}^3 \approx 1080 \text{ kg/m}^3$
Dry rover mass	m_{dry}	650 kg
Loaded rover mass	m_{loaded}	990 kg
Lunar gravity	g_{Moon}	1.62 m/s^2
Earth gravity	g_{Earth}	9.81 m/s^2
Applied design load factor	N_d	2.0

The bed volume implied by the locked CAD dimensions is

$$V_{\text{bed}} = L_{\text{bed}} B_{\text{bed}} H_{\text{bed}} = (1.20 \text{ m})(0.75 \text{ m})(0.35 \text{ m}) = 0.315 \text{ m}^3. \quad (1)$$

The corresponding effective bulk density is

$$\rho_{\text{reg}} = \frac{m_p}{V_{\text{bed}}} = \frac{340 \text{ kg}}{0.315 \text{ m}^3} = 1079.4 \text{ kg/m}^3 \approx 1080 \text{ kg/m}^3. \quad (2)$$

Consistency note. Week 4 first rounded the volume to 0.30 m^3 , producing approximately 1133 kg/m^3 , but the same Week 4 report later calculated the exact CAD volume as 0.315 m^3 [2]. Week 5 used 0.315 m^3 and 1079 kg/m^3 [3]. This Week 7 report carries forward the exact CAD volume and the Week 5 density. Similarly, the loaded mass remains 990 kg; 1000 kg is used only as a rounded planning value.

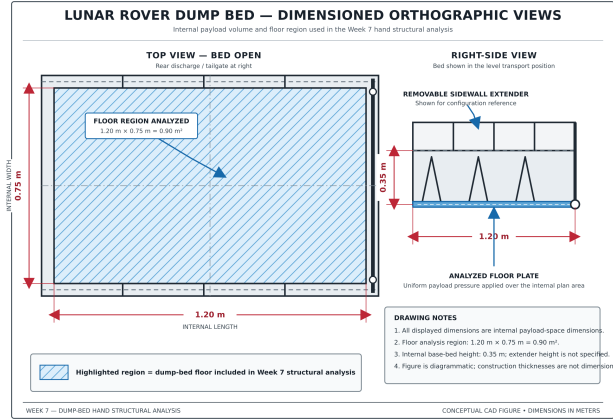


Figure 2: Dimensioned dump-bed geometry used to calculate floor area, internal volume, and uniform payload pressure.

The selected Week 4 geometry is also consistent with the Week 3 preliminary ranges. Week 3 proposed an overall length of 2.5 m to 3.0 m, overall width of 1.5 m to 2.0 m, and wheel diameter of 0.45 m to 0.65 m [1]. Week 4 then selected approximately 2.8 m overall length, 1.8 m track width, and 0.50 m wheel diameter [2]. No geometric conflict is created by carrying those values into Week 7.

4.2 Provisional material

The first calculation assumes an unstiffened 6061-T6 aluminum parent-metal sheet. The values are provisional conceptual properties based on published aluminum data [5, 6] and must later be replaced by minimum certified properties for the purchased product form and by appropriate joint or weld allowables.

Table 2: Provisional 6061-T6 aluminum properties used in the hand calculation.

Property	Symbol	Assumed value	Use
Elastic modulus	E	68.9 GPa	Beam deflection
Tensile yield strength	σ_y	276 MPa	Parent-metal yield check
Density	ρ_{Al}	2700 kg/m ³	Floor mass
Poisson's ratio	ν	0.33	Recorded for plate/FEA work; not used in the strip model

Kaiser Aluminum lists typical 6061-T6/T651 yield strength of 276 MPa and elastic modulus of 68.3 GPa [5]. This report retains the commonly used conceptual value $E = 68.9$ GPa so the Week 7 calculations remain consistent with the first report version. The difference is approximately 0.9 percent and does not change the design recommendation. The 276 MPa value is a typical parent-metal property, not a welded-joint allowable. NASA aluminum-welding guidance identifies reduced strength in heat-affected zones as an important joint consideration [7].

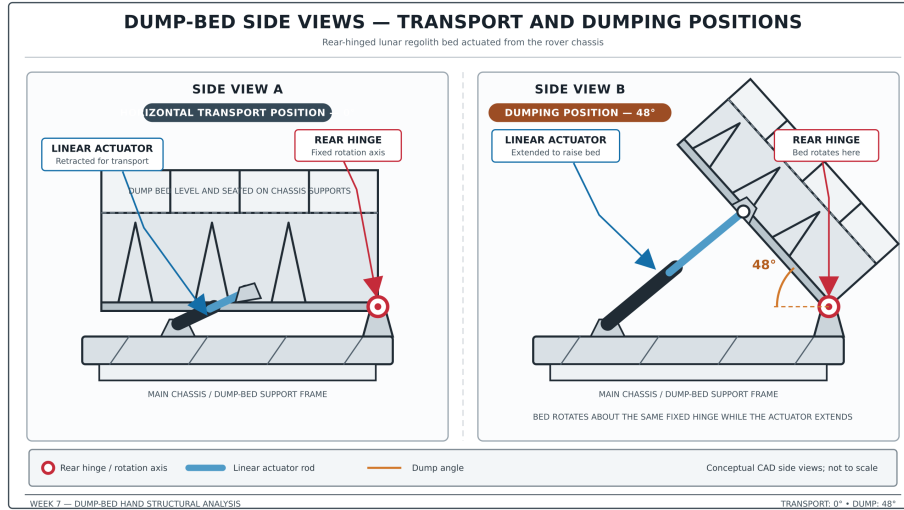


Figure 3: Rear-hinged dump-bed arrangement and actuator-assisted unloading geometry carried forward from the Week 3 concept.

5 Loading Cases

Table 3: Uniform payload load cases used for the floor-sheet comparison.

Case	Gravity	Payload force	Nominal pressure	Factor	Design pressure
Lunar operation	1.62 m/s ²	550.8 N	612 Pa	2.0	1224 Pa
Earth handling/test	9.81 m/s ²	3335.4 N	3706 Pa	2.0 sensitivity	7412 Pa

The 2.0 design factor is carried forward from the Week 5 mobility convention [3]. The Earth case is not the normal mission condition. It represents shop handling, ground testing, demonstrations, and accidental loading of the bed before launch. Because Earth gravity is

$$\frac{g_{\text{Earth}}}{g_{\text{Moon}}} = \frac{9.81 \text{ m/s}^2}{1.62 \text{ m/s}^2} = 6.06 \quad (3)$$

times lunar gravity, Earth handling can govern stiffness even though the rover operates on the Moon.

6 Free-Body Diagram Description

The bed floor is treated as a continuous sheet supported by transverse cross-members. Between adjacent cross-members, a 1.0 m-wide strip of floor is isolated. The strip is replaced by a simply supported beam of span L_s and thickness t . Uniform pressure p becomes a line load $q = p(1.0 \text{ m})$. The standard simply supported, uniformly loaded beam relationships used in this report follow mechanics-of-materials beam theory [8]. The support reactions are equal because the load is uniform

and symmetric:

$$R_A = R_B = \frac{qL_s}{2}. \quad (4)$$

The maximum positive bending moment and maximum vertical deflection occur at midspan.

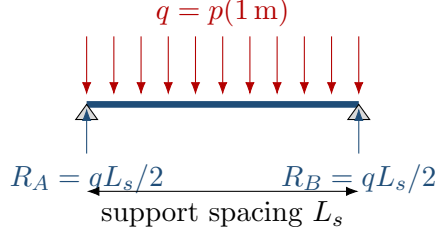


Figure 4: Free-body diagram of the one-meter-wide dump-bed floor strip between adjacent cross-members.

The supports are idealized as pins or rollers. A welded or bolted sheet may have partial rotational restraint and two-way plate action, which would generally lower the global midspan moment. The simply supported strip is conservative for global one-way bending, but it does not capture local stresses near welds, fasteners, edges, or concentrated loads.

7 Lunar Payload Load

The lunar payload force is

$$F_{p,\text{Moon}} = m_p g_{\text{Moon}}. \quad (5)$$

Substituting the locked payload and lunar gravity,

$$F_{p,\text{Moon}} = (340 \text{ kg})(1.62 \text{ m/s}^2) = 550.8 \text{ kg}\cdot\text{m/s}^2 = 550.8 \text{ N}. \quad (6)$$

8 Uniform Floor Pressure

The floor area is

$$A = L_{\text{bed}} B_{\text{bed}} = (1.20 \text{ m})(0.75 \text{ m}) = 0.90 \text{ m}^2. \quad (7)$$

The nominal lunar pressure is

$$p_{\text{Moon}} = \frac{F_{p,\text{Moon}}}{A} = \frac{550.8 \text{ N}}{0.90 \text{ m}^2} = 612 \text{ N/m}^2 = 612 \text{ Pa}. \quad (8)$$

Applying the same 2.0 design factor used in the Week 5 mobility analysis,

$$p_{\text{Moon},d} = N_d p_{\text{Moon}} = (2.0)(612 \text{ Pa}) = 1224 \text{ Pa}. \quad (9)$$

9 Structural Model and Boundary Conditions

This report uses a conservative simply supported beam approximation, not a rectangular plate equation. The floor strip has width $b = 1.0$ m, thickness t , and support span L_s . The uniform line load is

$$q = pb. \quad (10)$$

For a rectangular strip, the second moment of area is

$$I = \frac{bt^3}{12}, \quad (11)$$

and the section modulus is

$$S = \frac{I}{t/2} = \frac{bt^2}{6}. \quad (12)$$

The one-meter strip is a per-unit-width model. Although the physical bed is only 0.75 m wide, stress and deflection in this uniform strip model are independent of the selected strip width because both the line load and the strip section properties scale with b . Cross-member cases of 0.30 m, 0.40 m, and 0.60 m represent longitudinal floor spans. The 0.75 m case separately represents a conservative side-to-side strip spanning between bed side supports with no effective intermediate support.

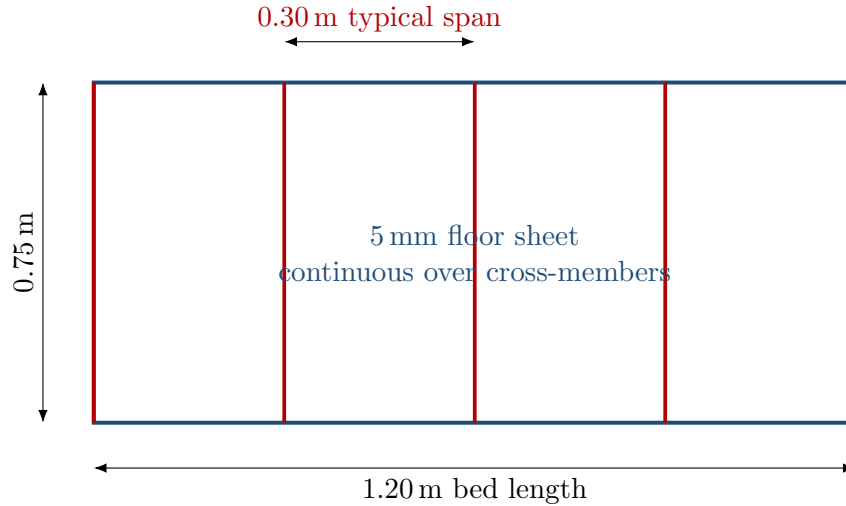


Figure 5: Recommended floor layout with cross-member support lines at 0.30 m centers.

10 Maximum Shear and Bending Moment

For a simply supported beam with a uniform line load [8],

$$V_{\max} = \frac{qL_s}{2}, \quad (13)$$

$$M_{\max} = \frac{qL_s^2}{8}. \quad (14)$$

For the recommended 5 mm sheet and 0.30 m support spacing under lunar design pressure,

$$q_{\text{Moon},d} = p_{\text{Moon},d}b = (1224 \text{ N/m}^2)(1.0 \text{ m}) = 1224 \text{ N/m}. \quad (15)$$

Therefore,

$$V_{\text{max}} = \frac{(1224 \text{ N/m})(0.30 \text{ m})}{2} = 183.6 \text{ N}, \quad (16)$$

and

$$M_{\text{max}} = \frac{(1224 \text{ N/m})(0.30 \text{ m})^2}{8} = 13.77 \text{ N.m}. \quad (17)$$

11 Bending Stress Calculation

For $b = 1.0 \text{ m}$ and $t = 0.005 \text{ m}$,

$$I = \frac{(1.0 \text{ m})(0.005 \text{ m})^3}{12} = 1.0417 \times 10^{-8} \text{ m}^4. \quad (18)$$

The section modulus is

$$S = \frac{I}{t/2} = \frac{1.0417 \times 10^{-8} \text{ m}^4}{0.0025 \text{ m}} = 4.1667 \times 10^{-6} \text{ m}^3. \quad (19)$$

The maximum bending stress is

$$\sigma_{\text{max}} = \frac{M_{\text{max}}}{S}, \quad (20)$$

so

$$\sigma_{\text{max},d} = \frac{13.77 \text{ N.m}}{4.1667 \times 10^{-6} \text{ m}^3} = 3.3048 \text{ e6 Pa} = 3.305 \text{ MPa}. \quad (21)$$

Pass: 3.305 MPa is far below the provisional 276 MPa parent-metal yield strength. This does not validate the complete floor because local impact, joints, hinge loads, and cross-member stresses are not represented.

12 Deflection Calculation

For a simply supported beam under uniform line load [8],

$$\delta_{\text{max}} = \frac{5qL_s^4}{384EI}. \quad (22)$$

Substitution gives

$$\delta_{\text{max},d} = \frac{5(1224 \text{ N/m})(0.30 \text{ m})^4}{384(68.9 \times 10^9 \text{ N/m}^2)(1.0417 \times 10^{-8} \text{ m}^4)} = 1.7987 \times 10^{-4} \text{ m} = 0.180 \text{ mm}. \quad (23)$$

A preliminary serviceability criterion of $L_s/200$ is used:

$$\delta_{\text{allow}} = \frac{L_s}{200} = \frac{0.30 \text{ m}}{200} = 0.00150 \text{ m} = 1.50 \text{ mm}. \quad (24)$$

Pass: 0.180 mm is less than the assumed 1.50 mm limit. The $L/200$ limit is a conceptual comparison criterion, not a lunar-rover design-code requirement.

13 Factor of Safety

The yield factor of safety is

$$\text{FoS}_y = \frac{\sigma_y}{\sigma_{\max}}. \quad (25)$$

At the factored lunar design load,

$$\text{FoS}_{y,d} = \frac{276 \text{ MPa}}{3.305 \text{ MPa}} = 83.5. \quad (26)$$

At nominal lunar pressure, the calculated stress is 1.652 MPa, giving

$$\text{FoS}_{y,\text{nom}} = \frac{276 \text{ MPa}}{1.652 \text{ MPa}} = 167.0. \quad (27)$$

The applied 2.0 load factor has already doubled the pressure before the design stress was calculated. A design-load factor of safety above 1.0 therefore passes the stated 2.0 load-factor requirement. The additional ratio of 83.5 is only the remaining theoretical parent-metal yield margin for the ideal uniform strip. It is not the factor of safety of the complete dump-bed assembly.

14 Thickness or Cross-Member Sensitivity Study

Table 4 applies the same lunar design pressure of 1224 Pa to every candidate. Pass/fail uses both parent-metal yielding and the assumed $L_s/200$ deflection limit. All candidates pass yielding. The widest, thinnest cases fail because deflection increases with L_s^4 and decreases with t^3 .

Table 4: Lunar design-load beam-strip sensitivity for $p = 1224 \text{ Pa}$ and $E = 68.9 \text{ GPa}$.

L_s (m)	t (mm)	$\sigma_{\max}$ (MPa)	$\delta_{\max}$ (mm)	FoS_y	Result
0.30	4	5.164	0.351	53.4	PASS
0.30	5	3.305	0.180	83.5	PASS
0.30	6	2.295	0.104	120.3	PASS
0.30	8	1.291	0.044	213.8	PASS
0.40	4	9.180	1.110	30.1	PASS
0.40	5	5.875	0.568	47.0	PASS
0.40	6	4.080	0.329	67.6	PASS
0.40	8	2.295	0.139	120.3	PASS
0.60	4	20.655	5.621	13.4	FAIL: $\delta > 3.00 \text{ mm}$
0.60	5	13.219	2.878	20.9	PASS
0.60	6	9.180	1.665	30.1	PASS
0.60	8	5.164	0.703	53.4	PASS
0.75	4	32.273	13.723	8.6	FAIL: $\delta > 3.75 \text{ mm}$
0.75	5	20.655	7.026	13.4	FAIL: $\delta > 3.75 \text{ mm}$

L_s (m)	t (mm)	$\sigma_{\max}$ (MPa)	$\delta_{\max}$ (mm)	FoS _y	Result
0.75	6	14.344	4.066	19.2	FAIL: $\delta > 3.75$ mm
0.75	8	8.068	1.715	34.2	PASS

14.1 Floor mass comparison

The solid sheet mass is

$$m_{\text{floor}} = \rho_{\text{Al}} L_{\text{bed}} B_{\text{bed}} t. \quad (28)$$

Table 5: Solid floor-sheet mass comparison for the 1.20 m by 0.75 m bed floor.

Thickness	Sheet mass	Mass saved vs. 30 mm	Comment
4 mm	9.72 kg	63.18 kg	Lightest, but limited Earth design stiffness
5 mm	12.15 kg	60.75 kg	Recommended with 0.30 m supports
6 mm	14.58 kg	58.32 kg	Acceptable with 0.40 m supports
8 mm	19.44 kg	53.46 kg	Unnecessarily heavy with close supports
30 mm	72.90 kg	—	Block-CAD placeholder; excessive as a solid floor

For the recommended sheet,

$$m_{\text{floor},5\text{ mm}} = (2700 \text{ kg/m}^3)(1.20 \text{ m})(0.75 \text{ m})(0.005 \text{ m}) = 12.15 \text{ kg}. \quad (29)$$

For the original 30 mm block-CAD placeholder,

$$m_{\text{floor},30\text{ mm}} = 72.90 \text{ kg}, \quad (30)$$

so the sheet-only mass reduction is

$$\Delta m = 72.90 \text{ kg} - 12.15 \text{ kg} = 60.75 \text{ kg}. \quad (31)$$

15 Earth-Gravity Handling Sensitivity

The Earth payload force is

$$F_{p,\text{Earth}} = m_p g_{\text{Earth}} = (340 \text{ kg})(9.81 \text{ m/s}^2) = 3335.4 \text{ N}. \quad (32)$$

The nominal Earth pressure is

$$p_{\text{Earth}} = \frac{3335.4 \text{ N}}{0.90 \text{ m}^2} = 3706 \text{ Pa}. \quad (33)$$

Applying a 2.0 ground-handling sensitivity factor,

$$p_{\text{Earth},d} = (2.0)(3706 \text{ Pa}) = 7412 \text{ Pa}. \quad (34)$$

Earth design pressure is $7412/1224 = 6.06$ times the lunar design pressure. Stress and deflection scale linearly with pressure in this elastic model, so Earth handling is the governing uniform-load sensitivity.

Table 6: Earth design-load sensitivity for $p = 7412 \text{ Pa}$ and an $L_s/200$ deflection criterion.

L_s (m)	t (mm)	$\sigma_{\max}$ (MPa)	$\delta_{\max}$ (mm)	FoS _y	Result
0.30	4	31.269	2.127	8.8	FAIL: $\delta > 1.50 \text{ mm}$
0.30	5	20.012	1.089	13.8	PASS
0.30	6	13.898	0.630	19.9	PASS
0.30	8	7.817	0.266	35.3	PASS
0.40	4	55.590	6.724	5.0	FAIL: $\delta > 2.00 \text{ mm}$
0.40	5	35.578	3.442	7.8	FAIL: $\delta > 2.00 \text{ mm}$
0.40	6	24.707	1.992	11.2	PASS
0.40	8	13.898	0.840	19.9	PASS
0.60	4	125.078	34.038	2.2	FAIL: $\delta > 3.00 \text{ mm}$
0.60	5	80.050	17.427	3.4	FAIL: $\delta > 3.00 \text{ mm}$
0.60	6	55.590	10.085	5.0	FAIL: $\delta > 3.00 \text{ mm}$
0.60	8	31.269	4.255	8.8	FAIL: $\delta > 3.00 \text{ mm}$
0.75	4	195.434	83.100	1.4	FAIL: $\delta > 3.75 \text{ mm}$
0.75	5	125.078	42.547	2.2	FAIL: $\delta > 3.75 \text{ mm}$
0.75	6	86.859	24.622	3.2	FAIL: $\delta > 3.75 \text{ mm}$
0.75	8	48.858	10.388	5.6	FAIL: $\delta > 3.75 \text{ mm}$

15.1 Recommended configuration under Earth handling

For $t = 5 \text{ mm}$ and $L_s = 0.30 \text{ m}$,

$$M_{\max,\text{Earth},d} = \frac{(7412 \text{ N/m})(0.30 \text{ m})^2}{8} = 83.385 \text{ N.m}, \quad (35)$$

$$\sigma_{\max,\text{Earth},d} = \frac{83.385 \text{ N.m}}{4.1667 \times 10^{-6} \text{ m}^3} = 20.012 \text{ MPa}, \quad (36)$$

$$\delta_{\max,\text{Earth},d} = 1.089 \text{ mm} < \frac{L_s}{200} = 1.50 \text{ mm}, \quad (37)$$

and

$$\text{FoS}_{y,\text{Earth},d} = \frac{276 \text{ MPa}}{20.012 \text{ MPa}} = 13.79. \quad (38)$$

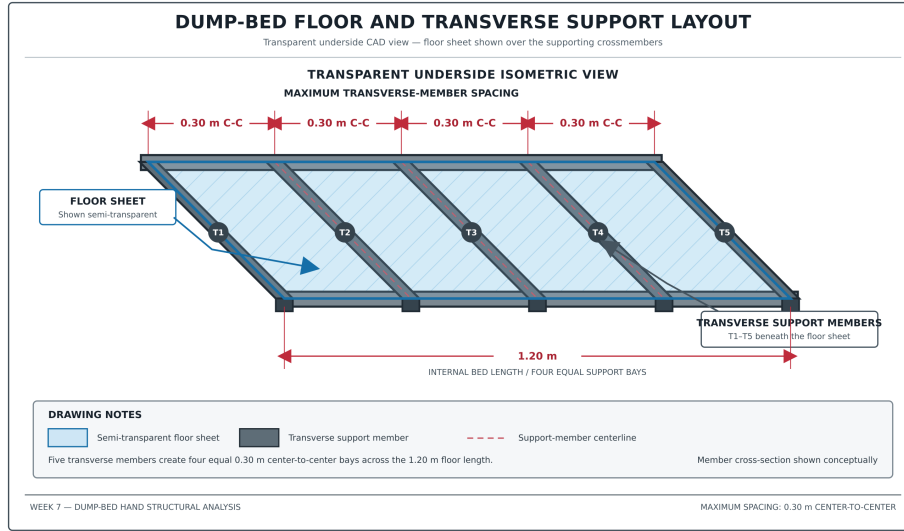


Figure 6: Recommended 5 mm dump-bed floor supported by transverse cross-members at no more than 0.30 m centers.

Pass: The selected configuration satisfies both the yield check and the assumed $L/200$ stiffness check at twice the static Earth payload. At nominal Earth gravity, the calculated stress is 10.006 MPa and the deflection is 0.545 mm.

A 6 mm sheet at 0.40 m spacing also passes, but its Earth design deflection is 1.992 mm against a 2.00 mm limit. This leaves almost no stiffness margin for imperfect support, wear, manufacturing tolerance, or localized loading. The 5 mm/0.30 m arrangement is preferred because it is lighter and has a more robust stiffness margin, provided the additional cross-member is practical.

16 Chassis Check

This section is a sanity check only. It does not assume that the Week 6 100 mm by 50 mm suspension beam is the chassis rail. A separate provisional chassis rail is proposed: 150 mm high by 75 mm wide by 5 mm wall 6061-T6 rectangular hollow section, with the 150 mm dimension vertical. Two rails span 2.20 m and share the loaded rover weight equally. The total lunar weight is multiplied by the same 2.0 design factor.

Using the exact loaded mass carried forward from Week 5,

$$W_{\text{rover}} = m_{\text{loaded}} g_{\text{Moon}} = (990 \text{ kg})(1.62 \text{ m/s}^2) = 1603.8 \text{ N}. \quad (39)$$

The factored total rover weight is

$$W_{\text{rover},d} = (2.0)(1603.8 \text{ N}) = 3207.6 \text{ N}. \quad (40)$$

The factored load per rail is

$$W_{\text{rail},d} = \frac{W_{\text{rover},d}}{2} = 1603.8 \text{ N}. \quad (41)$$

For a 2.20 m span, the equivalent uniform line load is

$$w_{\text{rail}} = \frac{W_{\text{rail},d}}{L} = \frac{1603.8 \text{ N}}{2.20 \text{ m}} = 729.0 \text{ N/m}. \quad (42)$$

For an RHS with outside width B , outside height H , and wall thickness t ,

$$I_{\text{rail}} = \frac{BH^3 - (B - 2t)(H - 2t)^3}{12}. \quad (43)$$

Substituting $B = 0.075 \text{ m}$, $H = 0.150 \text{ m}$, and $t = 0.005 \text{ m}$,

$$I_{\text{rail}} = \frac{0.075(0.150)^3 - 0.065(0.140)^3}{12} = 6.2304 \times 10^{-6} \text{ m}^4. \quad (44)$$

The rail section modulus is

$$S_{\text{rail}} = \frac{I_{\text{rail}}}{H/2} = \frac{6.2304 \times 10^{-6} \text{ m}^4}{0.075 \text{ m}} = 8.3072 \times 10^{-5} \text{ m}^3. \quad (45)$$

The maximum rail shear and bending moment are

$$V_{\text{max,rail}} = \frac{wL}{2} = \frac{(729.0 \text{ N/m})(2.20 \text{ m})}{2} = 801.9 \text{ N}, \quad (46)$$

$$M_{\text{max,rail}} = \frac{wL^2}{8} = \frac{(729.0 \text{ N/m})(2.20 \text{ m})^2}{8} = 441.0 \text{ N.m}. \quad (47)$$

The maximum rail stress is

$$\sigma_{\text{max,rail}} = \frac{M_{\text{max,rail}}}{S_{\text{rail}}} = \frac{441.0 \text{ N.m}}{8.3072 \times 10^{-5} \text{ m}^3} = 5.31 \text{ MPa}. \quad (48)$$

The maximum rail deflection is

$$\delta_{\text{max,rail}} = \frac{5wL^4}{384EI} = 0.518 \text{ mm}, \quad (49)$$

and the parent-metal yield factor of safety is

$$\text{FoS}_{y,\text{rail}} = \frac{276 \text{ MPa}}{5.31 \text{ MPa}} = 52.0. \quad (50)$$

The proposed rail passes this ideal global bending check. Two 2.20 m rails would have a combined bare-section mass of approximately 25.5 kg. However, the high calculated factor of safety indicates that uniform global bending is probably not the governing chassis mode. Suspension reactions, bed hinge and actuator loads, battery mounts, cross-rail torsion, welded joints, local wall buckling, and rollover or twist cases must be added before this rail size is finalized.

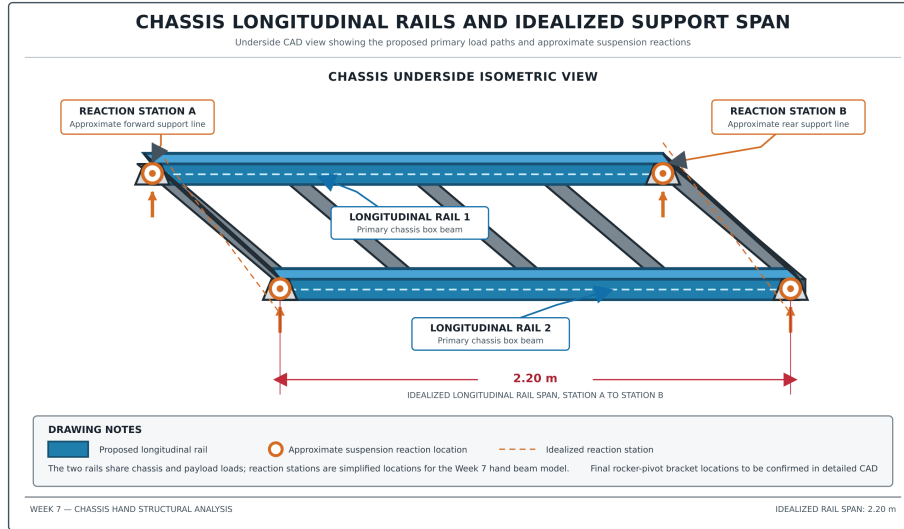


Figure 7: Provisional two-rail chassis model used only for the global bending sanity check.

17 Design Recommendation

The recommended preliminary dump-bed floor geometry is:

- **Floor:** 5 mm 6061-T6 aluminum sheet, subject to final product-form and joint allowables.
- **Plan size:** 1.20 m by 0.75 m inside the bed.
- **Supports:** transverse cross-members at a maximum of 0.30 m center-to-center spacing along the 1.20 m bed length. A practical layout places support lines at 0, 0.30, 0.60, 0.90, and 1.20 m, using the end frame as support where possible.
- **Lunar design result:** $\sigma_{\max} = 3.305 \text{ MPa}$, $\delta_{\max} = 0.180 \text{ mm}$, and $\text{FoS}_y = 83.5$ relative to typical parent-metal yield.
- **Earth design sensitivity:** $\sigma_{\max} = 20.012 \text{ MPa}$, $\delta_{\max} = 1.089 \text{ mm}$, and $\text{FoS}_y = 13.79$.
- **Estimated sheet mass:** 12.15 kg, which is 60.75 kg below the 30 mm solid-floor placeholder.
- Cross-members, side rails, hinge beam, actuator attachment, welds, and fasteners require separate sizing. This floor-sheet result does not size those parts.

A 4 mm floor is adequate for the ideal lunar uniform load at 0.30 m to 0.40 m support spacing, but it misses the assumed Earth design $L/200$ criterion at 0.30 m and would be more sensitive to dents and concentrated loading. A 6 mm floor at 0.40 m spacing is a reasonable alternative if fewer cross-members are required. The 8 mm sheet is unnecessarily heavy for a close-support layout. The 30 mm placeholder should not be retained as a solid aluminum floor.

For better mass efficiency than a thick solid plate, the preferred architecture is a thin sheet supported by cross-members or hat-section stiffeners. Sandwich construction can increase structural efficiency by separating strong faces with a low-density core, although joints, edges, inserts, and local damage require additional design attention [9]. For this conceptual rover, a 5 mm sheet with conventional cross-members is the clearest baseline because its construction and load path are easier to define at

the present design stage.

18 Limitations

- Uniform regolith pressure was assumed. Real loading may be heaped, asymmetric, impacted from a bucket, or concentrated by rocks.
- The floor strip was analyzed as a one-way simply supported beam. Two-way plate action, membrane action, and partial rotational restraint were omitted.
- The cross-members were treated as rigid supports and were not sized for bending, shear, torsion, weld load, or attachment load.
- The bed side walls, rear gate, hinge beam, linear-actuator mount, and load redistribution during a 45–50 degree dump were not analyzed.
- The horizontal loaded-bed position was used for global floor bending. Tilting reduces the gravity component normal to the floor but shifts regolith toward the hinge and rear region, creating local and asymmetric loads.
- Parent-metal 6061-T6 properties were used. Weld heat-affected-zone softening, minimum specified properties, temperature effects, lunar vacuum, radiation, abrasive regolith, and thermal cycling require separate treatment.
- Small-deflection linear elasticity was assumed. Local yielding, contact, buckling, joint slip, and geometric nonlinearity were not modeled.
- Dynamic loads from rover motion, drop loading, dump-bed slam, launch restraint, and emergency stops were not calculated. The factor of 2.0 is a conceptual static load factor, not a substitute for a defined load spectrum.
- Fatigue and crack growth were not assessed. Aluminum alloys may not show a clearly defined fatigue limit, so a static yield check cannot establish cyclic life [10].
- The chassis check assumes equal rail loading and simple end supports. Actual six-wheel rocker-bogie reactions and component point loads must replace that model.

19 Engineering Summary

The dump-bed floor is the correct primary Week 7 component because its 340 kg payload and 1.20 m by 0.75 m loaded area are defined. Under lunar gravity, the payload produces 550.8 N and a nominal floor pressure of 612 Pa. Applying the required factor of 2.0 gives a design pressure of 1224 Pa.

Using a conservative one-meter-wide simply supported beam strip, a 5 mm 6061-T6 floor spanning 0.30 m between cross-members develops a maximum lunar design bending moment of 13.77 N.m, maximum bending stress of 3.305 MPa, and maximum deflection of 0.180 mm. The calculated parent-metal yield factor of safety at the factored load is 83.5. Under a separate twice-static Earth handling case, the stress increases to 20.012 MPa and the deflection increases to 1.089 mm, which passes the assumed $L/200$ stiffness limit of 1.50 mm.

The recommended preliminary geometry is therefore a 5 mm aluminum floor with cross-members at no more than 0.30 m centers. The sheet mass is approximately 12.15 kg, compared with 72.90 kg for the 30 mm CAD placeholder. The 30 mm solid floor is not justified. This remains a preliminary conceptual hand analysis rather than validation of the complete dump-bed assembly.

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